Fulton Science Math Tournament

October 2026 - Individual Round

Honor Pledge

As a participant of FSMT, I promise to act with honesty, integrity, and respect in every challenge and competition I face. I will not seek or offer unauthorized assistance, and I will ensure that all work I submit is truly my own. I understand that fairness and trust are essential to the spirit of mathematics, and I accept full responsibility for upholding these values. I acknowledge that any breach of this pledge may lead to disqualification, and I am committed to competing with honor, diligence, and respect for my fellow competitors.

Full Name _____ Date _____

School _____ Grade _____

Team Name _____

PLEASE DO NOT BEGIN UNTIL BEING INSTRUCTED TO DO SO

Each student will independently tackle a set of 25 problems that are intentionally unrelated, designed to test a wide range of topics and difficulty levels. Participants will have a strict time limit of 40 minutes to complete all problems, emphasizing both speed and accuracy under pressure. For each problem, a correct answer is awarded 1 point, allowing each student to earn up to 25 points. Collectively, the team can achieve a maximum total of 100 points from this round.

It is important to note that in every written round of the competition, the answer must include the unit specified in the answer blank. These units are always presented in their plural form, regardless of whether a singular form might seem more appropriate based on the context of the problem. Answers that fail to include the exact unit as indicated will not be accepted. For additional clarification and a comprehensive explanation of rules, procedures, and scoring, participants should consult the Rules and Format Page provided. If you are still uncertain about a question, an answer, or the required format, ask a proctor for clarification immediately.

Individual Score _____ **Scorer's Initials** _____

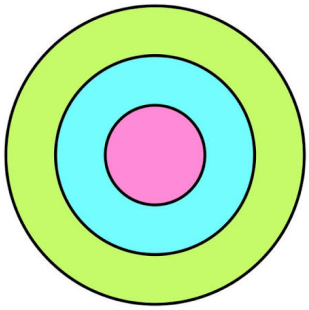
1. _____ What is 2026 divided by 2?

2. _____ What is the tens digit of 321×321?

3. _____ cups Albert is making orange juice with his super-secret recipe, which requires 3 oranges per cup. If he has 10 oranges, how many full cups of orange juice can he make?

4. _____ hearts Prayag is playing Minecraft on his pink computer. In Minecraft, the first 3 blocks of a fall cause no damage. After that, a player takes 1 heart of damage for every additional 2 blocks fallen. If Prayag falls 21 blocks, how many hearts of damage does he take?

5. _____ ways Billy wants to color in this shape so that every section is a different color. He has 3 different colors: green, blue, and pink. If he wants the center to be pink, how many ways are there to finish coloring the shape?



6. _____ A square has the same perimeter as a triangle with equal sides. If the triangle has side length 8, what is the side length of the square?

7. _____ Compute the product: $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6} \times 720$.

8. _____ Cal is walking up a flight of stairs. He walks up 4 floors, then walks down 1 floor, and finally walks up 6 floors. He ends up on the 14th floor. What floor did Cal start on?

9. _____ children A teacher wants to divide 24 identical pencils among several children so that every child gets a different positive number of pencils. What is the greatest possible number of children?

10. _____ If we define $a \bigstar b = 2b - a$, compute $(3 \bigstar 6) \bigstar (8 \bigstar 4)$.

11. _____ An elevator has buttons labeled 1 through 30. Ava notices that the buttons in the elevator behave weirdly and do not move to the desired floor. If the button has a prime number, the elevator moves up 1 floor. If the button number is divisible by 5, it moves up 3 floors. Otherwise, it moves up 2 floors. Ava begins on the first floor and presses 2, 6, 10, 11, 15, 17, 20. After pressing these buttons, which floor does she end up on?

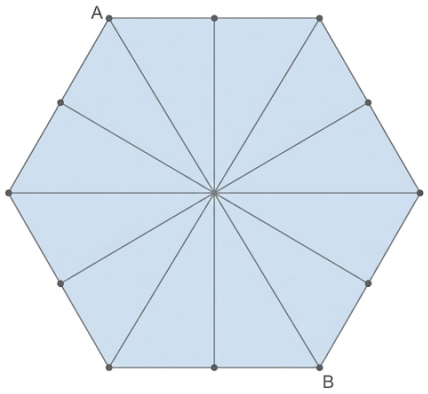
12. _____ The answer to this question is $x + 3$. Compute $\frac{x}{2}$.

13. _____ feet Prayag wants to use a ladder to reach the top of his house from the ground. His ladder is 13 feet long. Prayag sets the base of the ladder 5 feet away from his house and lays the top of the ladder against his house. The top of the ladder reaches the top of his house. How tall, in feet, is his house?

14. _____ pencils Freya has less than 100 pencils. When she puts them in groups of 2, she has 1 left over. When she puts them in groups of 5, she has 2 left over. If she puts them in groups of 7, she has 5 left over. How many pencils does Freya have?

15. _____ ways

The following diagram shows a regular hexagon with each vertex and the midpoint of each side connected to the center of the hexagon. If a person starts at vertex A and must travel to vertex B such that their straight-line distance to B strictly decreases with every single move, how many unique paths can they take? Vertices include all 6 corners, the 6 edge midpoints, and the center hub.



Each move must follow a drawn segment between adjacent vertices.

16. _____ cards

Alice, Bob, Charlie, and David are forming two ping-pong teams using 4 cards to determine the teams: 2 red and 2 black. Players who draw the same color are on the same team. Alice and Bob draw first. If their cards do not determine the teams, Charlie draws one of the remaining cards. What is the expected number of cards drawn?

17. _____ dollars

Jimmy is painting the 1-inch by 1-inch floor tiles in his room. On each tile, he paints a blue square whose vertices touch the midpoints of the tile. A gallon of Jimmy's favorite blue paint, Blubbering Bob's Blustering Blue Paint, can fully cover 144 square feet of floor tiles and costs $\$3.00$ per gallon. If the tiles in Jimmy's room cover a rectangular area of 20 feet by 30 feet, how much money will it cost Jimmy to paint? Assume that Jimmy can buy paint in any quantity at the listed price.

18. _____ Triangular pyramid $ABCD$ has E, F, and G on segments $\overline{AB}$, $\overline{AC}$, and $\overline{AD}$, respectively, such that $\frac{AE}{AB} = \frac{AF}{AC} = \frac{AG}{AD} = \frac{3}{4}$. The angle bisectors of triangle BCD intersect at point H. The ratio of the volumes of $ABCD$ and $EFGH$ is $\frac{x}{y}$, where x and y are relatively prime. Compute $x - y$.

19. _____ meters Jimmy and his brother Johnny went to the carnival together but got separated. Jimmy is at the hot dog stand while Johnny is at the face painting stand, and the distance between them is 56 meters. The two brothers agree to meet up, but due to a miscommunication, both Jimmy and Johnny end up walking to the other's stand. Since they weren't paying attention, they did not notice when they passed each other 32 meters away from the hot dog stand. After reaching their destination, each brother realizes their mistake and heads immediately back. This time, the brothers are paying attention and finally find each other. If both brothers walked at a constant speed the entire time, what distance from the hot dog stand would they find each other the second time?

They begin walking at the same time along the straight path connecting the stands.

20. _____ bites Zachary is eating a scrumptious mulberry pie. He is extremely hungry, so he eats half of the pie in one bite. However, he very quickly gets full from eating mulberry pie so each bite after that he eats half as much pie as the previous bite. How many more bites does he need to finish at least 99% of the pie?

21. _____ John, Jacob, Jingleheimer, and Schmidt are sitting around a circular table. Jingleheimer is sitting across from Schmidt, and John is sitting across from Jacob. Each person makes a claim:

- John states that Schmidt is sitting to his right.
- Jacob states that Jingleheimer is sitting to his left.
- Jingleheimer states that John is sitting to his right.
- Schmidt states that Jacob is sitting to his right.

What is the sum of all possible values of the number of people telling the truth?

22. _____ What is the sum of all the numbers of the form $\frac{1}{2^a \cdot 5^b \cdot 7^c}$ where a, b, and c are non-negative integers?

23. _____ A prime number p is a full-period prime if the decimal expansion of $\frac{1}{p}$ has a repeating sequence of length $p-1$. For example, 7 is the smallest full-period prime because $\frac{1}{7} = 0.\overline{142857}$, where the 6-digit sequence repeats indefinitely. What is the second smallest full-period prime?

24. _____ A frog is hopping along a number line stretching from 0 to ∞. The frog starts at 0 and every minute he either hops 2 or 3 units forward with equal probability. What is the probability that the frog will land on 8 at any point during his journey?

25. _____ Jack has 2025 empty boxes numbered 1-2025, as well as an infinite number of balls. Each time Jack eats a sandwich, he checks to see if any box has more than 4 balls in it. If so, he empties that box and puts a ball in the next box. Otherwise, he puts a ball in the first box. How many balls are in the boxes after Jack eats 480 sandwiches?